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20 February 1987
SUPERSEDING
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MILITARY SPECIFICATION

TRUCK, REFUSE COLLECTION; COMPACTION TYPE, SIDE LOADING, 16 CUBIC YARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven, four-wheel, two-rear-wheel drive, commercial refuse collection truck having a minimum gross vehicle weight (gvw) of 24,000 pounds (lbs). The truck shall be equipped with a stationary side loading, rear dumping body with a hydraulically compacting and ejecting plate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage, Vehicular, Ignition, Lighting and Starting.
ZZ-T-381 - Tires, Pneumatic, Vehicular (Highway).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2320

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STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications and standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B117 - Standard Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall be a four-wheel, two-rear-wheel drive, diesel or gasoline engine driven, cab and chassis equipped with a refuse collection body. The truck shall have a gvw rating of not less than 19,500 lbs. The refuse collection body shall be of the side loading type, and a hydraulically operated compacting and rear dumping ejecting plate system.

3.2 First production vehicle. The contractor shall furnish a truck for first production vehicle inspection.

3.3 Standard truck, components, and accessories. Except as specified herein, the truck, components, and accessories shall be the manufacturer's standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material,

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identical to that furnished to authorized company representatives for selection of trucks models and components. Truck, components, assemblies, and accessories to be provided hereunder are standard or optional commercial products for the purpose of this specification. Such items shall be construed as commercial motor vehicle products, and shall comply with all DoT Federal Motor Vehicle Safety Standards and Regulations applicable on the date of manufacture. The truck chassis model furnished shall not be older than manufacturer's current model on date of invitation to bid.

3.3.1 Drain plugs. Drain plugs installed in manual transmission and rear axle shall be of the permanent magnet type.

3.3.2 Heavy-duty cooling system. A heavy-duty cooling system shall be furnished that shall maintain engine coolant at a temperature below the boiling point with truck loaded to rated gvw and operated at an altitude of 10,000 feet above sea level, or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.3.3 Coolant system and indicators. Coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than 2-quart capacity. A low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the cab dash instrument panel.

3.3.4 Towing devices. Not less than two hooks or loops for towing the truck shall be furnished on the front of the truck.

3.3.5 Splash guards and shields. Rear wheels shall have rigid splash shields ahead of the wheels and splash guards to the rear. Splash and stone throw protection shall be in accordance with SAE J682. A metal strip, not less than 1/8-inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prelude bolt heads or bolt nuts from damaging the mud flap.

3.3.6 Air pollution control. Trucks shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, trucks destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.3.7 Sound level. The interior sound level shall conform to DoT Federal Motor Carrier Safety Regulation, section 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment Medium and Heavy Trucks.

3.3.8 Radio interference suppression. The vehicle shall be suppressed to limit the electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

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3.3.9 Engine hour meter. An engine hour meter, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the truck chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Truck weights, ratings, and dimensions.

3.5.1 Curb weight. Curb weight shall include the weight of the chassis and cab, with all attachments, accessories, equipment, body, full complement of fuel, lubricant, and coolant.

3.5.2 Weight rating. The gvw rating shall consist of curb weight, operator (weight computed at 175 lbs), and a payload of not less than 10,000 lbs.

3.5.3 Weight distribution. The distribution of the gvw for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.5.4 Ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. Minimum gvw shall be 19,500 lbs.

3.5.5 Overall width and height. The overall width of the vehicle exclusive of mirrors, lights, reflectors and tires shall be not more than 96 inches. Width over tires shall be not more than 100 inches. The overall height shall not be more than 140 inches.

3.5.6 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

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3.6 Performance.

3.6.1 Speeds and gradeability. High and low speed requirements shall be met with truck loaded to specified gvw.

3.6.1.1 High speed gradeability. The truck shall ascend a continuous grade of 1.5 percent at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive.

3.6.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall provide a truck speed of not more than 5 mph.

3.6.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Engine governed speed}}{\text{Total geared reduction X tire factor}}$$

3.6.2 Brake performance. Service brakes shall comply with the performance requirements specified in DoT Federal Motor Carrier Safety Regulation, section 393.52. Service brakes shall control and hold the truck, loaded to the rated gvw, on a 30 percent grade. Parking brake shall hold the truck, loaded to the rated gvw, headed either up or down the grade, without slipping on a 30 percent grade.

3.7 Chassis components.

3.7.1 Engine. Engine furnished shall be the chassis manufacturer's standard, or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

3.7.1.1 Diesel engine. Unless otherwise specified vehicles shall be equipped with a liquid cooled, compression ignitions, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower shall be determined in accordance with SAE J816 or J1349.

3.7.1.2 Gasoline engine. When specified (see 6.2), gasoline engine shall be furnished and shall be the chassis manufacturer's standard, or optional engine for the commercial model truck which meets or exceeds the requirements of this specification. The truck shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at engine speed not more than manufacturer's recommended operating speed. The engine shall be capable of continuous, warranted operation on unleaded fuel. The rated net torque output of the engine shall be not less than 265 pound-feet. Engine net torque ratings and net horsepower shall be determined in accordance with SAE J1349 or J816.

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3.7.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.7.1.4 Governor. An engine governor shall be furnished and set and sealed to limit engine manufacturer's maximum recommended operating speed.

3.7.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided for diesel engine driven vehicles. Control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters, or complete thermostatic control of all coolant flow through the radiator.

3.7.2 Electrical systems. Electrical systems shall be in accordance with DoT Federal Motor Carrier System Regulations, section 393.27 through 393.31 and 393.33.

3.7.2.1 Ignition systems. For gasoline engine driven vehicles a 12-volt (V) direct current (dc) ignition system shall be furnished. Alternator of not less than 40-ampere (A) rated capacity and which provides not less than 16A dc output at normal engine idle speed shall be furnished.

3.7.2.2 Starting system. For diesel engine driven vehicles, a 12 or 24V dc starting system, with 12V lighting system, and not less than a 60A alternator, shall be furnished. Engine starting equipment shall include an ether starting system or a glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type, controlled from the driver's compartment, with a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

3.7.2.3 Lighting. All vehicle lights, reflectors, and wiring shall be as specified herein and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19, 393.20, and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails, unless recessed and fully protected, or mounted on vehicle bumpers.

3.7.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Vehicle shall be provided with a double-faced front signal units and single-faced rear signal units installed in accordance with SAE J588. Turn signal operating units shall have visible and audible flash indicator.

3.7.2.5 Batteries. Each battery shall be 12V potential. The total reserve capacity rating and total cold cranking ampere rating at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table I. Batteries shall be of the maintenance free type having maintenance free characteristics listed in W-B-131.

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TABLE I. Batteries.

Engine type	Reserve capacity (minute)	Cold cranking (amperes)
Diesel engine (less than 220 gross horsepower)	320	1200
Diesel engine (220 gross horsepower and over)	480	1740
Gasoline engine	100	450

3.7.3 Fuel system. Fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.65 and 393.67.

3.7.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil bath type, oil capacity shall be not less than 1 quart.

3.7.3.2 Fuel tanks. Fuel tank shall be manufacturer's standard with a total capacity of not less than 40 gallons. When more than one tank is furnished on a gasoline engine driven vehicle a selector valve connecting either tank to engine fuel intake shall be provided, and means shall be provided to monitor the fuel level of either tank from a single fuel gage. When more than one tank is furnished on a diesel engine driven vehicle, means shall be provided to assure equalized fuel level in both tanks.

3.7.3.3 Fuel and water separator. When specified (see 6.2), a fuel and water separator shall be furnished for diesel engine driven vehicles. The separator shall include a water coalescer, a see-through bowl and a drain valve.

3.7.4 Exhaust system. The exhaust system shall conform to DoT Federal Motor Carrier Safety Regulation, section 393.83. Vertical exhaust systems capable of being reached easily by personnel entering or leaving either side of the truck cab shall be provided with a heat shield. Stacked exhaust sytems shall be provided with a hinged rain cap.

3.7.4.1 Exhaust system safety shield. The exhaust system or the hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.7.5 Transmission. Unless otherwise specified (see 6.2), an automatic or semiautomatic transmission shall be provided. When specified (see 6.2), a manually shifted transmission shall be provided. Input torque capacity of

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transmission shall be at least equal to the maximum torque delivered by the engine. Transmission and axle gear ratio shall be selected to provide performance specified in 3.6.1.

3.7.5.1 Automatic or semiautomatic transmission. The automatic or semiautomatic transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with SAE J704 power take off opening(s). Heavy-duty driven gears for the power takeoff shall be provided. A caution plate reading: "DO NOT OPERATE VEHICLE WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.7.5.2 Manually shifted transmission. The manual transmission shall have not less than five forward speeds and one reverse speed. Transmission shall provide for synchronized shift for at least the two highest forward speeds. Transmission shall be provided with SAE J704 left and right power takeoff openings.

3.7.5.2.1 Clutch. Clutch shall be the largest capacity clutch offered for the truck gvw rating and engine furnished, with torque capacity exceeding maximum delivered engine torque.

3.7.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine as developed through the maximum gear train reduction.

3.7.7 Frame. The chassis frame shall be the manufacturer's standard, reinforced frame, or a heavy-duty frame, for the type and class vehicle furnished. The reinforced frame shall have frame reinforcement extending at least from the rear of the front suspension rear hanger bracket, to the front of the rear spring. Reinforcement in the form of fishplating is not acceptable. Both the reinforced frame and the heavy duty frame shall have sufficient resisting bending moment to provide the structural strength required to at least equal the loads imposed with the vehicle loaded to provide specified gvw.

3.7.8 Suspension. Chassis shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with truck loaded to gvw. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Chassis shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

3.7.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with truck loaded to gvw. A single reduction rear axle shall be furnished.

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3.7.10 Wheels, rims, tires, and tubes. Truck shall be equipped with single front and dual rear wheels. Rim and tire ratings shall conform to TRA recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on each truck. All tires shall be of the same load range. When specified (see 6.2), disk type wheels shall be furnished.

3.7.10.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at the wheel, at the ground, with truck loaded to rated gvw. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall conform to TRA recommendations or to ZZ-T-381 with a size designation system the same as the TRA.

3.7.10.2 Tubes. Tubes, when furnished, shall be of heavy-duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided in accordance with TRA recommendations.

3.7.10.3 Tire chain clearance. Clearance shall be provided for installation and use of tire chains in accordance with SAE J683. Allowance for spring deflection shall be included.

3.7.10.4 Spare wheel assembly. When specified (see 6.2), a spare wheel or rim shall be mounted on a carrier and installed in a readily accessible location on the truck. Carrier design shall enable removal or mounting of spare wheel assembly using only the tools specified in 3.7.17.1.

3.7.10.5 Spare tire. When specified (see 6.2), an inflated spare tire mounted on the spare wheel rim shall be furnished. The spare tire shall be of the same size, tread design, and load range as the other tires furnished on the truck.

3.7.11 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52.

3.7.11.1 Air brakes. The vehicle shall be equipped with all full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

a. Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7-1/4 cubic feet per minute.
b. Air storage reservoir(s) with not less than 2,000 cubic inches total capacity; each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank and check valves between compressor and last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air control valves.
e. Air pressure gage, visible to the driver.

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f. Low air pressure warning, visible and audible.
g. Service brake stop lamp switch.
h. Automatic moisture ejector.
i. When specified (see 6.2), an air dryer shall be furnished. The air dryer shall be of the desiccant type.
j. Automatic slack adjusters.

3.7.12 Cab. Manufacturer shall furnish a standard full-width cab. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with at least the curb side door equipped with external, key-operated lock. Drip rails shall be furnished above cab doors. Cab shall have upholstered, full-width, adjustable seat and back or individual, adjustable, driver's seat and individual passenger seat. The color of the upholstery and the interior finish shall be compatible with the exterior color. White upholstery shall not be furnished. Interior lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be furnished and installed.

3.7.13 Steering. Manufacturer's standard power steering shall be furnished.

3.7.14 Windshield wipers and washers. Truck shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by electric motor(s).

3.7.15 Bumper. Unless the bumper is an integral part of the truck cab, channel type front bumper shall be provided.

3.7.16 Rear end protection. Rear end of truck shall be protected in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.86.

3.7.17 Tool stowage. Stowage space of sufficient size to accommodate vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of cab, it shall be weatherproof and shall provide for locking with a padlock.

3.7.17.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging mounted tire assembly with the spare assembly, and shall include, as a minimum, a hydraulic jack, jack handle, and wheelnut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

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3.7.18 Heater and defroster. Hot water heater shall be provided. Heater shall have air intakes. Discharge outlets shall be provided to direct air to floor and to defroster louvers. Heater shall be complete with blower and mounted controls convenient to the driver.

3.7.19 Controls and operating mechanisms. Unless otherwise specified (see 6.2), controls and operating mechanisms shall be located for left-hand drive. When specified (see 6.2), the truck shall be provided with left-hand and right-hand drive controls and operating mechanisms. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

3.7.20 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Key-operated ignition switch.
b. Ammeter, charging indicator or voltmeter.
c. Fuel gage.
d. Oil pressure gage or indicator.
e. Speedometer with recording odometer.
f. Engine temperature gage or indicator.
g. Dual sunvisors.
h. Front door mounted armrest on driver's side.
i. Driver's compartment ventilator other than window.
j. Manufacturer's standard electric horn.
k. Tachometer for diesel engine driven vehicles.

3.7.21 Outside rearview mirrors. Combination (flat and convex) outside rearview mirrors, enclosed in common housing shall be provided on each side of cab. Each mirror shall have at least three supporting arms. Each combination mirror shall have at least 50 square inches of flat reflective area and convex surface having at least 20 square inches of reflective area.

3.8 Body components.

3.8.1 Refuse body construction. The refuse body shall be of the stationary, enclosed, power compaction type, fabricated by welding steel sheets and steel reinforcing structural members to make a sturdy and durable construction. The body shall be the manufacturer's standard barrel or box shape. All exterior box bracing shall be continuously welded to prevent bleeding of rust from inside the members onto the exterior of the vehicle. The design and reinforcement of the body shall withstand continuous operation at maximum imposed loads without evidence of deformation or excessive wear to the body and its equipment. The interior of the body shall have smooth easily cleanable surfaces, and shall be free of pockets and restrictions that will retard movement of refuse during loading and unloading cycles. The interior

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of the body shall be painted with a rust-inhibiting primer. All refuse material, including ashes and saturated garbage, shall be retained in the body without spillage or liquid leakage during operations. The entire body shall be liquid-tight to a height of not less than four inches above the floor.

3.8.1.1 Capacity. The body shall have compaction volume of not less than 16 cubic yards, including the volume forward of the rear edge of the side charging doors. The hopper capacity shall be equal to: the inside width of the body, times the width of the side door opening, times the distance from the lower edge of the side door to the body floor; shall be not less than 3/4 cubic yard, exclusive of the volume of the hopper occupied by the compaction plate with the plate in its foward most position.

3.8.1.2 Hopper charging doors. Each side of the body shall be provided with a hopper charging door located in the forward portion. Each hopper charging door shall provide an unobstructed opening not less than 36 inches wide and not less than 35 inches high. The height from ground level to the lower edge of the hopper charging door opening shall be the minimum required by the hopper capacity requirements of 3.8.1.1. The openings shall be enclosed with sliding or hinged doors. Latches to hold all doors closed during transit or during compaction operations shall be provided. Safety switches shall be provided to prevent movement of the compacting and ejection plate when hopper charging doors are open. Door openings shall be such that the contents of 55-gallon refuse containers can be easily dumped into the body. The edges around the openings shall be reinforced to resist denting.

3.8.1.3 Rear discharge door(s). The top or side hinged rear discharge door(s) shall, when open, provide an unrestricted opening for ejecting the refuse. The door(s) shall be reinforced to prevent deformation which may cause leaks between door(s) and body during packing operation. Latches shall be provided for holding the door(s) securely in locked position. Latch actuating lever(s) shall be readily operable by a man standing on the ground. The hinges shall keep the door(s) tightly alined to prevent leakage. Door(s) shall open and close without excessive effort by one operator. When a single top-hinged door is furnished, a hydraulic cylinder to open the door and keep it in a near horizontal plane, while body is being unloaded, shall be provided.

3.8.1.4 Packer plate. The packer plate shall be actuated by double acting telescopic hydraulic cylinder assembly and shall travel from the forward end of the hopper to the rear discharge door(s). The compacting side of the plate shall be shaped so as to cause the refuse to be compacted and forced rearward and upward during the compaction process. The packer plate shall closely fit the cross sectional dimensions of the compacting area to preclude bypassing the refuse and shall be guided to prevent scraping and gouging of the body inside walls. No permanent misalignment shall occur when maximum compaction pressure is applied to or released from the packer plate.

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3.8.1.5 Packer plate operation. The packer plate operation, for compacting and ejecting refuse, shall conform to the following:

a. The complete compacting cycle, comprised of the packer plate movement from the initial station at the loading hopper to any intermediate position and return, shall be accomplished in not more than 70 seconds.
b. The complete ejection cycle, comprised of the packer plate movement from the initial station at the loading hopper to the final location at the discharge door opening and return, shall be achieved in not more than 75 seconds.
c. The packer plate shall be capable of supplying a maximum compacting pressure of not less than 11.5 lbs per square inch over the effective packer plate area at the standard relief valve setting. The compacting plate driving system shall incorporate provisions to limit the compacting pressure to the manufacturer's maximum allowable compacting pressure.
d. The packer plate shall completely expel the total volume (maximum rated payload) of refuse in one pass without external assistance (manual racking or pulling of body contents).

3.8.1.6 Sump tank. A liquid collection tank of not less than 12-gallon capacity shall be provided under the body to catch drainage from wet refuse. Drain shall be provided to drain the sump tank. Sump tank need not be provided when the same provision is incorporated in the refuse body.

3.8.1.7 Retaining teeth. Body shall have at least two rows of retaining teeth on the top inside to prevent refuse from falling forward after compaction stroke.

3.8.2 Hydraulic system. The hydraulic system shall be complete with oil reservoir, fluid level gage, filtration circuit compatible with SAE J931, pump, control and relief valves, and all other components required to actuate and control the hydraulically operated components. All surfaces of hydraulic ram piston rods shall be chrome plated the full length with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B117. The connection ends shall be treated in the the best commercial practice. Manual lever or electric pushbutton type controls shall be provided for controlling packer plate movement, operable from both sides of the body near the hopper charging doors. Provisions for adjusting the degree of engine acceleration during packing operations shall be furnished. Electric pushbutton type controls shall have means to de-energize the control system to prevent accidental initiation of packer plate movement. When electric type controls are furnished, a manually operated hydraulic system shall be properly labeled. Overload devices, set to limit the pressures to the manufacturer's recommended maximum operating pressures, shall be incorporated in all hydraulic power equipment. The hydraulic oil reservoir shall have adequate capacity to hold the maximum oil volume for any combination of attachments plus not less than 20 percent of additional volume. A pressure gage shall be provided to measure hydraulic pressure and shall be permanently installed to

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indicate the operating pressure. When specified (see 6.2), the electric controls shall allow completely automatic cycling of the packer plate at the option of the operator.

3.8.3 Power takeoff. Power to hydraulic pump shall be transmitted by a heavy-duty power takeoff and drive line. All operating mechanisms shall be capable of withstanding not less than 100 percent overload. Power takeoff shall be engaged and disengaged from the truck cab.

3.8.4 Steps and signals. A nonskid riding step and grab handles shall be provided at each hopper charging door for loading personnel to ride between collection points. A signal bell or buzzer shall be installed in the truck cab, audible to the driver, and operable by pushbuttons installed adjacent to the hopper charging doors.

3.8.5 Fenders. Rear fenders shall be provided on the truck when cylndrical body design is furnished.

3.9 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.10 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.11 Rustproofing. When specified (see 6.2), the truck cab, chassis, and chassis underside shall be rustproofed in accordance with FED-STD-297.

3.12 Servicing and adjusting. Prior to acceptance of the truck by the Government, the contractor shall service and adjust the truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.

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c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment.
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.13 Identification marking. Identification shall be permanently and legibly marked directly on the truck or on a corrosion-resisting metal plate securely attached to the truck at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.14 Workmanship.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.14.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity with the requirements of this contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions under the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight, the operator weight of 175 lbs, and the payload (see 3.5.2 and 3.5.3), to provide the specified gvw. The calculated imposed loads on front and rear axle will be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet requirements of 3.7.8, 3.7.9, and 3.7.10.1.

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4.3.2 Road test. The truck loaded with all the equipment specified herein, and with payload of refuse, shall be driven at speeds up to 55 mph over improved roads for not less than 25 miles. At least five sudden stops shall be made from a speed of not less than 20 mph.

4.3.3 Loading and compaction test. Truck shall be loaded, with normally collected refuse, with the specified payload (see 3.5.2). Compaction members, hydraulic system, hopper charging door movements, and controls shall be observed during operations. Time required for compaction cycles shall be observed to verify conformance to 3.8.1.5a and c. During the loading test, the packer plate shall be operated through a total of 25 compactions and return cycles as follows:

a. Five cycles after compaction of first hopper load.
b. Five cycles after filling the body to 1/4 of the load volume.
c. Five cycles after filling the body to 1/2 of the load volume.
d. Five cycles after filling the body to 3/4 of the load volume.
e. Five cycles after filling the body to the full load volume.

4.3.3.1 Gross vehicle weight. The loaded truck shall be weighed upon completion of loading and compaction to provide the gvw specified in 3.5.3.

4.3.4 Dumping test. The loaded truck shall be driven to a dumping area and the load discharged to determine that discharge door(s) operation conform to 3.8.1.3. Time used for complete ejection cycle and the ejection of the load of refuse shall be observed to determine conformance to 3.8.1.5b and d.

4.3.5 Seal test. After completion of the foregoing tests, truck shall be loaded with water to a depth of four inches to determine seal fit and liquid-tight capabilities specified in 3.8.1.

4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.3.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last truck shipped on the contract. The first production vehicle shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first production vehicle in a serviceable condition for the duration of the contract.

4.4 Production truck tests. The contractor's testing system shall, as a minimum, assure that the truck is capable of meeting performance requirements contained herein. At the point of final acceptance the contractor shall make available to the Government records acceptable to the Government indicating that the servicing and adjusting required by 3.12 has been accomplished.

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4.5 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The complete truck shall be preserved in accordance with the contractor's standard practice. When specified (see 6.2), the complete truck shall be preserved and packaged in accordance with the mobile requirements of MIL-STD-281, with the level of preservation and packaging as specified (see 6.2).

6. NOTES

6.1 Intended use. The refuse collection truck covered by this specification is intended for use in loading, transporting, and dumping miscellaneous refuse generated at military installations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When gasoline engine is required (see 3.7.1.2).
c. When fuel and water separator is required (see 3.7.3.3).
d. When manual transmission is required (see 3.7.5).
e. When disk type wheels are required (see 3.7.10).
f. When bias ply tires are required (see 3.7.10.1).
g. When spare wheel or rim and carrier are required (see 3.7.10.4).
h. When spare tire is required (see 3.7.10.5).
i. When air dryer is required (see 3.7.11.1.i).
j. When tire change tools are required (see 3.7.17.1).
k. When left-hand and right-hand drive controls and operating mechanisms are required (see 3.7.19).
l. When completely automatic packer plate controls are required (see 3.8.2).
m. When color of paint finish coat is specified (see 3.10).
n. When rustproofing is required (see 3.11).
o. When preservation and packaging in accordance with MIL-STD-281 is required, and the level of preservation and packaging required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contract or in accordance with the contract or purchase order requirements.

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6.4 Subject term (key word) listing.

Compacting
Comression ignition engine
Hydraulic equipment
Refuse
Truck

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Review Activity:
Air Force - 84

Preparing Activity:
Navy - YD

(Project 2320-0291)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-18479G	TRUCK, REFUSE COLLECTION; COMPACTION TYPE, SIDE LOADING, 16 CUBIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
YARD	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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